

Loan Management System - Entity Relationship Design

Project Overview

This project is a **Loan Management System** that supports four user roles:

- **USER** – Registers, logs in, views loan products, applies for loans, and tracks loan status.
 - **UNDERWRITER** – Reviews loan applications and approves or rejects them.
 - **MANAGER** – Monitors loan approvals and overall loan portfolio.
 - **ADMIN** – Manages loan products, users, and system configurations.
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1. Customer Entity

Table Name

customers

Description

Stores customer profile information, authentication details, and system roles.

Primary Key

customer_id (Long)

Fields

Field	Type	Description
customer_id	Long	Primary Key
customer_name	String	Customer Name
email	String	Unique Email
password	String	Encrypted Password
phone	String	Mobile Number
address	String	Customer Address
branch	String	Branch Name

Field	Type	Description
role	Enum	USER / UNDERWRITER / MANAGER / ADMIN
status	Enum	ACTIVE / INACTIVE
created_at	LocalDateTime	Record Created Time
updated_at	LocalDateTime	Last Updated Time

Relationship

Customer (1)

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|

|----- (N)

LoanApplication

A customer can submit multiple loan applications.

2. Loan Product Entity

Table Name

loan_products

Description

Stores all loan schemes offered by the organization.

Primary Key

loan_code (String)

Fields

Field	Type	Description
loan_code	String	Primary Key
loan_name	String	Loan Name

Field	Type	Description
loan_type	Enum	PERSONAL / HOME / VEHICLE / EDUCATION / BUSINESS
minimum_amount	Decimal	Minimum Loan Amount
maximum_amount	Decimal	Maximum Loan Amount
interest_rate	Decimal	Annual Interest Rate
minimum_tenure	Integer	Minimum Tenure (Months)
maximum_tenure	Integer	Maximum Tenure (Months)
processing_fee	Decimal	Processing Fee
daily_penalty_rate	Decimal	Daily Penalty Rate
active	Boolean	Product Availability
created_at	LocalDateTime	Created Date
updated_at	LocalDateTime	Updated Date

Relationship

LoanProduct (1)

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|----- (N)

LoanApplication

One loan product can be selected by many customers.

3. Loan Application Entity

Table Name

loan_applications

Description

Represents a customer's request for a loan before approval.

Primary Key

application_id (Long)

Fields

Field	Type	Description
application_id	Long	Primary Key
customer_id	FK	Customer Reference
loan_code	FK	Loan Product Reference
requested_amount	Decimal	Requested Loan Amount
tenure_months	Integer	Requested Tenure
application_date	LocalDate	Date Applied
status	Enum	SUBMITTED / UNDER_REVIEW / APPROVED / REJECTED
remarks	String	Underwriter Remarks

Relationships

Many LoanApplications -> One Customer

Many LoanApplications -> One LoanProduct

4. Loan Account Entity

Table Name

loan_accounts

Description

Created only after a loan application is approved. Represents the sanctioned loan.

Primary Key

loan_account_id (Long)

Fields

Field	Type	Description
loan_account_id	Long	Primary Key
loan_number	String	Unique Loan Number
application_id	FK	Loan Application
approved_amount	Decimal	Approved Amount
interest_rate	Decimal	Final Interest Rate
tenure_months	Integer	Approved Tenure
emi_amount	Decimal	Monthly EMI
application_date	LocalDate	Application Date
approval_date	LocalDate	Approval Date
disbursement_date	LocalDate	Loan Disbursement Date
loan_start_date	LocalDate	Loan Start Date
next_emi_date	LocalDate	Next EMI Due Date
loan_close_date	LocalDate	Loan Closure Date
status	Enum	APPROVED / DISBURSED / ACTIVE / CLOSED

Relationship

LoanApplication (1)

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|----- (1)

LoanAccount

Each approved application results in one loan account.

5. EMI Payment Entity

Table Name

emi_payments

Description

Stores EMI payment transactions for each loan account.

Primary Key

payment_id (Long)

Fields

Field	Type	Description
payment_id	Long	Primary Key
loan_account_id	FK	Loan Account Reference
installment_no	Integer	EMI Number
due_date	LocalDate	EMI Due Date
payment_date	LocalDate	Payment Date
emi_amount	Decimal	Scheduled EMI
principal_paid	Decimal	Principal Component
interest_paid	Decimal	Interest Component
penalty_paid	Decimal	Penalty Amount
total_paid	Decimal	Total Amount Paid
payment_type	Enum	CASH / CARD / ONLINE / UPI

Field	Type	Description
status	Enum	PENDING / PAID / OVERDUE

Relationship

LoanAccount (1)

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|----- (N)

EmiPayment

A loan account contains multiple EMI payments.

Entity Relationships

Customer (1)

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|

|----- (N)

LoanApplication

|

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(N) ---|--- (1)

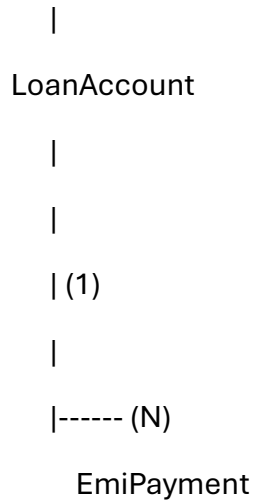
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LoanProduct

|

|

| (1)



Cardinality Summary

Parent Entity	Child Entity	Relationship
Customer	LoanApplication	One-to-Many
LoanProduct	LoanApplication	One-to-Many
LoanApplication	LoanAccount	One-to-One
LoanAccount	EmiPayment	One-to-Many

Business Workflow

Customer Registration/Login



View Available Loan Products



Select Loan Product



Submit Loan Application



Underwriter Reviews Application



Approve / Reject



If Approved



Loan Account Created



Loan Disbursed



Monthly EMI Payments



Loan Closed

Frontend Module Mapping

Customer Module

- Register
- Login
- Profile
- My Loan Applications
- My Loan Accounts
- EMI History

Loan Product Module

- View Loan Products
- Loan Product Details

Loan Application Module

- Apply for Loan
- Track Application Status
- View Underwriter Remarks

Loan Account Module

- View Active Loans
- Loan Details

- Outstanding Balance
- Next EMI Information

EMI Module

- Pay EMI
- Payment History
- Download Payment Receipt

Admin Module

- Manage Customers
- Manage Loan Products
- Manage Employees
- Dashboard

Underwriter Module

- View Pending Applications
- Review Customer Details
- Approve Loan
- Reject Loan
- Add Review Remarks

Manager Module

- View Approved Loans
- Loan Reports
- Portfolio Summary
- Approval Statistics